

Ali Zain

AI & Machine Learning Engineer

methealizain@gmail.com | +92 3229587927 | [LinkedIn](#) | [GitHub](#) | [Portfolio Website](#)

Professional Summary

AI/ML Engineer with 2 years of experience developing and deploying end-to-end LLM applications, Conversational Voice Agents, ML model training, RAG pipelines, and computer vision systems. Proficient in FastAPI, vector search indexing, and AWS cloud deployment. Track record of cutting manual workflows by **up to 80%** across legal, marketing, and fintech domains through structured data extraction, tool-calling agents, and custom object-detection models.

Work Experience

Machine Learning & Python Developer

Intalytic Group Ltd, Islamabad

November 2024 – March 2026

- Trained and optimized a **YOLOv8** person-detection model for squash-court analytics (annotation to evaluation) at **89% mAP@0.5**, and a Random Forest speech-confidence classifier at 84% accuracy.
- Built document processing pipeline and led API development for OCR, extraction, validation, and automated storage of banking documents to **AWS S3**, reducing manual verification workload by 60%.
- Developed an AI-powered timesheet-to-payroll pipeline for [payseye.com](#), migrating from GPT-4o to **GPT-5.4-mini** for structured data extraction and optimizing via **few-shot prompting** and chunking to reduce hallucination; deployed to AWS with a 4+ member team.

AI & Python Developer

Exact Solution Electronics, Poland

March 2026 – Present

- Built and deployed a production **outbound voice AI agent** (Twilio, ElevenLabs, FastAPI) that autonomously handles 200+ lead-qualification calls daily, with intent-recovery, retry, and human-handoff fallbacks, monitored via **AWS CloudWatch** logs and alerts.
- Automated post-call intelligence (transcription, multilingual translation, summaries, and lead-interest/intent classification), letting sales teams prioritize hot leads and cutting lead-review time by **25%**.
- Shipped a **multi-agent SEO blog pipeline** (LangGraph, Claude, GPT) that automates research, drafting, and keyword optimization, cutting manual blog writing effort by **~80%**.
- Designed an [AI sales agent](#) (OpenAI tool-calling, FastAPI, MongoDB) that converts natural-language queries into multi-attribute MongoDB aggregation pipelines and returns interactive product cards in **~2s avg latency**.

PROJECTS

Firmhole | Final Year Project

2024

- Built a SaaS platform for law firms that automates document workflows with summarization and classification, cutting case processing time by 20%.

EDUCATION

BSIT | Quaid-i-Azam University, Islamabad | CGPA 3.3/4.0

Graduated: 2020 - 2024

SKILLS & Certifications

AI/ML: Scikit-learn, TensorFlow, PyTorch, LangChain, LangGraph, LangSmith, RAG, YOLO, OpenAI, Hugging Face
Backend & Tools: Python, FastAPI, Twilio, ElevenLabs, Docker, AWS, PostgreSQL, MongoDB, Pgvector, Weaviate

[Machine Learning Specialization](#) | [Retrieval Augmented Generation](#) | [Microsoft Certified: Azure AI Fundamentals](#)